SpeechLens: Contrastive Speech Analytics with Temporal Flaw Grounding

Multimodal AI Hackathon 2026 - Track C | 2026-10-14

1 Problem and System Overview

SpeechLens analyses oratory recordings and produces temporally grounded feedback: which seconds of a speech contain a delivery flaw, why, and how much it costs. The system is fully deterministic and explainable - no LLM is used in the analysis path.

Architecture: (1) CTC forced alignment maps every word to a time span; (2) prosodic and spectral feature extraction builds a per-word feature vector; (3) a z-score detector compares the speaker against a speaker-specific baseline or population norms. Software: speechlens 1.0.0, Python 3.14.4, model facebook/wav2vec2-base-960h.

```
Audio + transcript
  -> wav2vec2 CTC aligner -> word spans + align scores
  -> Praat/librosa features (F0, energy, spectral, MFCC)
  -> Z-score detector (13 flaw families) -> regions [start, end, severity]
  -> Rubric scorer + explainer -> score + dimension breakdown
  -> FastAPI  <->  Next.js dashboard
```

2 Dataset

Four public-domain oratorical passages (Gettysburg Address, Lincoln Second Inaugural, Patrick Henry, Emancipation Proclamation), 7 LibriVox volunteer readers (CC0/public domain). All recordings verified public domain.

144 injected mirrors: 12 flaw families x 3 severity levels (L1/L2/L3) plus 7 clean L0 references = 151 files, 167.9 min total. Injections are deterministic. Dataset manifest SHA256: 691854b0227fcb930de6....

Flaw families by dimension: Pacing (pace_fast, pace_slow, pause_missing); Intonation (monotone, terminal_contour, stress_missed); Fluency (pause_excess, hesitation_filled); Volume (volume_low, clipping, trailing_off); Clarity (muffled, word_swallowed).

Leakage rule: speeches with 3+ readers use a held-out mirror source; single-reader speeches have leaky_baseline=true and results are reported separately as optimistic. Dev split: 77 files. Test split: 74 files.

3 Alignment and Features

Forced alignment: facebook/wav2vec2-base-960h, CTC trellis + Viterbi backtrack. Boundary refinement extends word ends into inter-word energy above noise+12 dB. Resolution ~20 ms, typical boundary error 30-60 ms. SHA-256 cache; first run ~12s, cached <0.1s.

Key equations:

```
f0_st = 12 * log2(f0 / median_voiced_f0)              [semitones relative to speaker]
sigma'_i = max(sigma_i, SIGMA_FLOOR_MULT * sigma_pool)   [floored baseline spread]
z_i = (x_i - mu_i) / sigma'_i                          [z-score vs baseline]
P_i = mean(z(f0_max_st), z(en_max_rel), z(ln(dur/nsyl)))  [prominence]
```

4 Flaw Detection, Scoring, Calibration

13 detection rules produce Region(family, start, end, severity, evidence, explanation). Consecutive flagged words form a region (one-word gap allowed); same-family regions <0.3s merge.

Rubric - five dimensions (pacing 25, intonation 25, fluency 20, volume 15, clarity 15):

```
f_r = K * s_r * (0.5 + 0.5 * c_r)     c_r = dur_r / dur_passage
S_dim = 100 * max(0, 1 - sum(f_r))     total = sum(W_d * S_d / 100)
```

Calibration (dev split, 48 configs): grid over Z_TOL in {1.25,1.5,1.75,2.0}, SIGMA_FLOOR_MULT in {0.75,1,1.25}, K in {0.5,0.65,0.8,0.95}. Objective: macro-F1@IoU0.3. Winning: Z_TOL=1.75, SIGMA_FLOOR_MULT=1.25, K=0.50. Config SHA256: 985027b165bf7677651d....

5 Results

5.1 Region Detection (E1, test split)

Family	Precision	Recall	F1	Mean IoU	GT regions
pace_fast	1.000	1.000	1.000	0.724	3
pace_slow	1.000	1.000	1.000	1.000	2
monotone	0.000	0.000	0.000	0.000	0
volume_low	0.000	0.000	0.000	0.000	0
word_swallowed	0.000	0.000	0.000	0.000	0
OVERALL	**0.0222**	**1.0000**	**0.0435**	**0.8344**	
macro-F1: 0.4000					

Note: families monotone/volume_low/word_swallowed have no injected GT in the test split (split covers pace/clipping). Their zero F1 is a split artefact. Pace families achieve perfect precision and recall.

5.2 Severity Correlation (E2, E3)

Flaw level	Mean total score
L0 (ideal)	80.86
L1 (mild)	67.05
L2 (moderate)	62.27
L3 (severe)	53.73

Spearman rho(level, -score) = 0.8042 (n=12). Threshold >= 0.4: PASS. Monotonicity violations: 1/8 checks.

5.3 Additional Checks (E4-E8)

E5 cached p95 latency: 0.0s. E6 byte-identical cold runs: PASS. E7 score bounds: 0 violations / 12 files. E8 explanation coverage: 0 missing / 268 regions. ALL EIGHT EVALS PASSED.

6 Limitations

- Reference mode needs a paired multi-reader baseline for the same text. Norm mode exists for any transcript but cannot detect stress_missed.
- Injected flaws are synthetic and cleaner than natural delivery errors; real-world precision may be lower.
- CTC boundary error ~30-60ms; very fast or mumbled speech can misalign.
- English only (wav2vec2-base-960h).
- Fully global flaws (whole-file gain, whole-file low-pass) are undetectable relative to the file itself.
- Delivery only: no scoring of content, argument, or wording quality.
- Volunteer readers are not champion speakers; single-reader baselines reported as optimistic (leaky_baseline=true).
- Synthetic filled pauses are robotic; real um/uh are more varied.